

Normal Sinus Rhythm

Patient: *Emily Carter, 24*

Presentation

Emily is brought in after working outdoors in 105°F heat. She reports dizziness and mild nausea.

- HR: 82 bpm
- BP: 110/70
- Rhythm: Regular

Monitor

Normal P wave → Narrow QRS → Consistent PR interval

Electrical Explanation

Impulse originates in the **Sinoatrial node**, travels through atria → **Atrioventricular node** → His-Purkinje system.

Cellular Level

- Normal Phase 4 depolarization in pacemaker cells
- Balanced Na^+ and Ca^{2+} influx

EMS Treatment

No antiarrhythmics indicated.

Treatment: IV fluids.

Teaching Purpose

Establish baseline conduction before pathology.